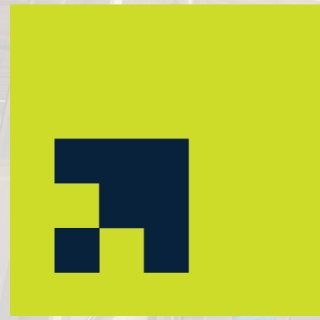




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Cross-Dataset Generalization for Network Intrusion Detection Using Supervised and Unsupervised Machine Learning Models

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Abstract

Network Intrusion has been on the rise ever since the invention of internet and IoT devices that specifically accumulate huge amounts of data. Network Intrusion Detection System (NIDS) has been a major topic considering the number of threats involved in today's cyber security scenario. With the ever-rising threats in communication networks, development of NIDS plays a boosting factor to prevent the networks from attacks of all kinds. The purpose of the development of network intrusion detection system is to protect the communication network, preserve and safeguard vital information from intruders dynamically by analyzing the complex and sophisticated traffic data. The motives for the intrusion vary on individual and institutional levels, financial gains, unauthorized access of intellectual properties, confidential information, political gains and others. NIDS monitors and analyzes all the incoming web traffics, detects them as 'Attack' or 'Benign' thus safeguarding the network from future attacks of same kinds. Several research has been carried out to detect network intrusion for specific fields (Health, Finance, Entertainment, Military, Government) incorporating advanced techniques with the utilization of specific data. Previously published research papers showed that the deployment of robust machine learning algorithm achieved greater performance in intrusion detection when trained and tested on the same dataset. Research portals such as JSRT Journal, MDPI, ResearchGate, ScienceDirect, Springer Nature, IEEE contain research papers related to cross dataset generalization (mostly between CICIDS2017 and CSE-CIC-IDS2018). Most of the research papers drew their conclusions based on 'Accuracy' of the models for intrusion detection. There was no uniformity in these papers. Some described the usage of SMOTE while training CICIDS2017 dataset in supervised learning models, variation on the size of dataset and percentage of data used for training the models; thus, leaving a gap for cross-dataset generalization to the unseen network traffic. This research paper tries to showcase the capabilities of Supervised Learning Models (Random Forest, XGBoost), Unsupervised Learning Models (PCA, K-Means, Autoencoders) and Deep Learning Models in cross-dataset generalization. All the models will be trained on CICIDS2017 dataset and tested on two different test datasets namely CSE-CIC-IDS2018 and UNSW-NB15; which are publicly available. No SMOTE will be applied to oversample the CICIDS2017 dataset. Unlike previous papers, this research will prioritize the evaluation metrics significant for cross-dataset generalization. Finally, the performance of the models will allow us to assess the strength and practicality of a unique approach in real-world cybersecurity scenario.

Declaration of Originality

This is for declaration of originality...

Dedication

This is for dedication...

Contents

Chapter 1

Introduction

The growth of internet has significantly assisted humans, organizations and government in ways no one ever dreamt of ages ago. Internet has simplified the lives of individuals and organizations. Development of robust communication platforms, enhanced economy across the globe, elevation in military operations, education, health, entertainment, science, research and invention are some of the privileges provided by the internet. Along with the privileges come the burdens in the form of cyberattacks. Intruders tend to overcome the traditional security firewalls and benefit by constantly bombarding malicious attacks that the obsolete web security tools and even antiviruses software like Kaspersky could never detect [?]. Thus, urges the need for the deployment of Network Intrusion Detection System (NIDS). Robust AI mechanism and machine learning algorithms assist NIDS to detect sophisticated unseen attacks. They help in identifying the patterns of the malicious attacks analyze them and blocks them enabling a secured network infrastructure in the future. The deployment of Supervised Machine Learning models such as Random Forest, XGBoost, Unsupervised Learning models (PCA, K- Means, Autoencoders) and Deep Learning models exhibit better performance in intrusion detection from unknown attacks with improved accuracy and F1- Score [?]; [?]; [?] particularly on CICIDS2017 and UNSW-NB15 datasets easily available on public domain. Unlike previous datasets such as KDD Cup 1999 and NSL – KDD, these datasets contain real network traffic with varied attacks inhibiting wider traffic patterns [?, ?]. Even though several AI driven approaches have been considered and developed to detect intrusion in the computer network with better accuracy, some models have notable performance on the benchmark datasets (CICIDS2017, CSE-CIC-IDS2018 and UNSW-NB15) alone but fail while deployed to detect unknown incoming traffic and thus underperform considering their capabilities in cross-dataset generalization across varied computer network. Intruders take an upper hand in this scenario by deploying more sophisticated versions of web attacks continuously [?]. Some of the reasons why these models fail to generalize conveniently.

- Class Imbalance – Most of the web traffics are ‘Genuine’ and few are actual ‘Attack’.
- Variable features and size – The training and testing datasets consist of different features; also, the size of the features sourced from different computer network alter.
- Computational power – Researchers at an individual level often find it difficult to evaluate the performance of the models given the size of the datasets and the computational power of the device.

These issues motivate the researchers to find alternative ways to build robust NIDS to safeguard the computer network from intruders. This allows researchers to learn new patterns of attacks and dynamically adjust the models from a classical approach to a revolutionary approach [?].

Chapter 2

Foundations

This is a chapter...

Chapter 3

Literature Review

Some of the research carried out on Cross-Dataset Generalization for NIDS have shown improvements with advanced dataset handling and feature engineering on supervised machine learning and deep learning models. These models have achieved greater accuracy on benchmark datasets and improved to some extent in cross-dataset generalization (between CICIDS2017 and CSE-CIC-IDS2018, UNSW-NB15). However, accuracy alone cannot be considered to determine how robust the model is given the high number of genuine network flows and a smaller number of attacks. Cross-Dataset Generalization for NIDS face reliability issue when models are deployed for intrusion detection considering the increasing number of attacks in heterogeneous environments with huge domain shifts when the models with greater accuracy on benchmark dataset fail to detect intrusion in an unknown dataset yielding more False Positives. Generalization becomes complex as the features from the train dataset that are predictive of malicious activities within the dataset may fail to predict malicious behavior from a different dataset thus dropping the performance of the models [?]. Uniformity in dataset size for training and testing, usage of SMOTE are crucial to make comparisons on how the models performed in intrusion detection in the future.

The studies have shown the effectiveness of ensemble learning and feature selection in finding nonlinear dependencies among different labels in the network flow by keeping low dimensionality. They have also cautioned to interpret the evaluation wisely despite Random Forest, XGBoost, 1D-CNN achieving accuracy 85% on benchmark datasets [?], these models faced challenges modelling on complex multi-stage attacks [?].

3.1 Cross-Dataset Generalization using Supervised Learning Models

With the ever-increasing study and experiments on network intrusion detection, a recent study by [?], although not validated and reviewed by experts performed cross-dataset generalization by employing machine learning algorithms on CIC-IDS-2017, CSE-CIC-IDS2018, LycoS-IDS2017, and LycoS-UnicasIDS2018 datasets and eventually applied corrections to CSE-CIC-IDS2018 based on the results of LycoS-IDS2017. The study revealed that the performance dropped to detect malicious behaviors for generalization in comparison to near perfect results in intrusion detection when trained and tested on same datasets. The supervised learning models learned bigger scale attack types such as DOS and DDoS and generalized well. Meanwhile, due to class imbalance of the attack types, the models failed to detect web attacks and infiltration attacks. The researchers have pointed out that the variability of the datasets, irregular distribution of attacks among datasets and redundancy in the samples played a causal role in performance drop of the models failing to learn the attack patterns. Problems associated with feature calculation, protocol identification, feature multiplication and labeling in CIC-IDS-2017 degraded the quality of the data and hence corrected with LycoS-IDS2017. The findings suggested that the learning of the transferrable attacks might not be the real reason, but the features associated with the dataset that hinder the performance of the models. As per the study, the models should be implemented with prior knowledge of different traffics that were never encountered before.

The table below showcases the performance of the models while trained on CIC-IDS-2017 and

tested on CSE-CIC-IDS2018 as per [?] for cross-dataset generalization for network intrusion detection.

Table 1: Model Evaluation Results from [?].

Train Dataset	Test Dataset	Models	Evaluation Metrics		
			MCC	F1	AUROC
CIC-IDS-2017	CSE-CIC-IDS2018	Random Forest	25.96%	25.76%	66.93%
		XGBoost	35.68%	26.43%	80.44%

Another study by [?] revealed the performance drop for NIDS while evaluated on recent datasets. Random Forest when trained on CIC-IDS-2017 and tested on CSE-CIC-IDS2018 saw a Macro F1 drop of 0.059 from baseline 0.941 to 0.882. Similarly, 1D-CNN saw a decline of 0.070 from 0.942 to 0.872. It seems obvious that the performance degrades while generalizing to a completely different dataset when both temporal and network domain distance increases implying that performance on benchmark dataset does not signify robustness for future attack detection. The study highlights the importance of carrying out temporal forecasting evaluation while implementing models for intrusion detection.

3.2 Hybrid Deep Learning approach for Cross-Dataset Generalization

In addition to the application of supervised learning models, usage of deep learning methods has been observed for the detection of intrusion due to its capability of learning hierarchical representations from the complex traffic data. A research paper “Real-Time Network Intrusion Detection System Using Deep Learning: Identifying Cyber Attacks Through Anomaly Pattern Recognition” [?] has properly analyzed the performance of LSTM, CNN and Autoencoders in detecting anomalies. The paper has discussed the hybrid approach and proposed the Real-Time NIDS by employing attention mechanism, LSTM, CNN and Autoencoders. In this hybrid architecture setup, LSTMs model network flow sequences, CNNs identify spatial patterns from the traffic features, Autoencoder calculates the anomaly score from the reconstruction error and lastly the important characteristics/temporal phases within the traffic data sequence are identified by attention mechanism.

Like the performance of the supervised learning models on benchmark datasets, application of deep learning for intrusion detection recorded exemplary F1 scores of 97.8%, 97.3% and 98.0% for CIC-IDS-2017, CSE-CIC-IDS2018 and UNSW-NB15 dataset respectively. The study has conducted cross-dataset performance between CIC-IDS-2017 and UNSW-NB15 only. Using this hybrid approach, they have yielded a F1-score of 93.5% and FPR of 2.8% only which has surpassed the performance of the base supervised learning models. The major difference in the results between the application of supervised learning models from previous research and the hybrid DNN approach lie in data split for training and testing. The study of cross-dataset generalization by [?] used whole dataset for training and testing. Meanwhile, the research paper by [?] used SMOTE (to oversample rare attack classes), 70% of each dataset for training, 15% for hyperparameter tuning and remaining 15% for testing. While this approach yielded superior performance, as per Elangovan et al. (2026), this complex deep learning architecture does not guarantee a reliable generalization as the performance for intrusion detection degrade when there are large distribution shifts and may not be resilient enough to identify unfamiliar traffic thus contradicting the notion that complex model would yield better performance when the models are trained on older traffics and tested on newer version of web traffics with evolving network technologies, communication behaviors and varied attack patterns. Despite

the robustness of the models, [?] has stated the necessity of preserving the network features when the network traffic features are classified based on their semantics during dataset shift. Protocol specific indicators showing higher degree of uncertainty whereas communication features like flow rate and temporal activity features generalizing well on different datasets. The negative communication behavior aspects that arise in different situations can be handled by certain behavioral characteristics. The evidence for this is discussed in the representation-drift study by [?]. Higher the difference between training and testing datasets with respect to time and field, higher the construction error. Models trained on CIC-IDS-2017 showed 0.0211 mean reconstruction error while analyzed on CSE-CIC-IDS2018, increased to 0.0342 for CICIoT2023 and reached the maximum of 0.0384 when applied to CIC-DIAD2024 showing the latent representations obtained from the original dataset diverged from the target-domain traffic.

The findings from the research for cross-dataset generalization by [?, ?, ?] are in contrast considering the models used. These papers show how the evaluation design, datasets, pre-processing, feature engineering and model design affect generalization. Thus, cross-dataset generalization for NIDS remains continuous deep research of the specific source and target datasets, features compatibility, attack overlap and experimental approach.

3.3 Research Gap

All these previous studies of Cross-Dataset generalization for NIDS using Supervised Machine Learning and Deep Learning Models demonstrate gradual progress for NIDS, although the evaluations were made using different metrics on different test datasets. Trees based models provide faster classification benchmarks while deep learning methods contribute to representation learning, temporal modelling, and anomaly detection. Hybrid deep learning method could significantly assist real-time cross-dataset generalization for NIDS. However, achieving accuracy greater than 80% or even 90% with high precision scores on a single benchmark dataset or a cross-dataset (as the results suggested by the studies above) does not confirm its robustness for deployment. They fail to exhibit excellent generalization capabilities. There is no consistent performance across varied test datasets thus failing to deliver similar results. The inconsistency in performance across datasets proves the significance of good labeling, dataset preparation and feature engineering. There remains a consistency gap in the advancement of NIDS evaluation approaches that would consider the efficiency of classification and generalization to different datasets, robustness over time, feature stability, imbalance class, false-positive estimations, and computational efficiency. Future research should conduct collaborative experiments for real deployment simulations with independent datasets and heterogeneous network conditions.

Chapter 4

Methodology

We have implemented a quantitative approach for the study of Cross-Dataset Generalization for Network Intrusion Detection Using Supervised and Unsupervised Machine Learning Models. The study focuses on the overall performance of the models in detection of 'Attack' from the test datasets by training the models on an entire train dataset. Since the focus of this study is to investigate the cross-dataset generalization capabilities of both supervised and unsupervised machine learning models for network intrusion detection, the models will be trained entirely on CIC-IDS-2017 dataset and evaluated on unseen CSE-CIC-IDS2018 and UNSW-NB15 datasets.

- **Research Type:** Quantitative
- **Data:** Publicly available datasets
 - **Train Dataset:** CIC-IDS-2017
 - **Test Dataset:** CSE-CIC-IDS2018 and UNSW-NB15
- **Sampling:** Non-Random Sampling
 - Convenience Sampling
- **Programming Language:** Python 3.11
- **Libraries:** Pandas, Numpy, Tensorflow, Keras, Scikit-learn, Optuna, RobustScaler, Matplotlib, Seaborn
- **Models:** Random Forest, XGBoost, DNN, Isolation Forest, One-Class SVM, Autoencoder
- **Evaluation Metrics:** F1-Score, MCC, ROC-AUC Curve

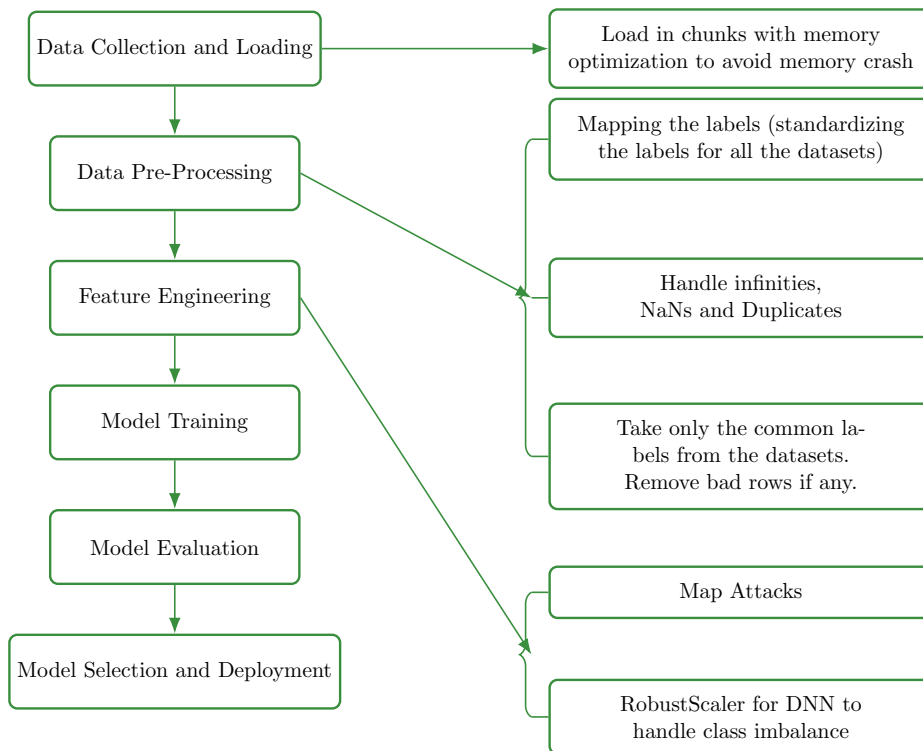


Figure 1: Workflow for Cross-Dataset Generalization for Network Intrusion Detection.

4.1 Dataset Collection and Loading

For the study of Cross-Dataset Generalization for Network Intrusion Detection Using Supervised and Unsupervised Machine Learning Models at an academic level, we have sourced three datasets from public domain. The collected data are then loaded in chunks with memory optimization (dowcasting some datatypes) to avoid memory crash.

Table 2: Dataset Details

Dataset	Source	Link for the Source	Dataset Size	Dataset Shape	Access Date
CIC-IDS-2017	Kaggle	https://www.kaggle.com/datasets/chethuhn/network-intrusion-dataset	884.65 MB	2,830,743, 79	22/06/2026
CSE-CIC-IDS2018	Kaggle	https://www.kaggle.com/datasets/solarmainframe/ids-intrusion-csv	6.89 GB	16,233,002, 84	22/06/2026
UNSW-NB15	Canadian Institute for Cybersecurity	https://cicresearch.ca/CIC-Dataset/CIC-UNSW/	1.78 GB	3,540,241, 84	22/06/2026

CIC-IDS-2017 and CSE- CIC-IDS2018 datasets were downloaded from Kaggle. Meanwhile, the UNSW-NB15 dataset was fetched from Canadian Institute for Cybersecurity portal by submitting necessary information. We will employ CIC-IDS-2017 as the train dataset, while CSE- CIC-IDS2018 and UNSW-NB15 will be our test datasets for cross-dataset generalization for NIDS. A quick exploration of these datasets before data pre-processing showed dominant presence of ‘Benign’ class in all the datasets. 2273097 (80.3%) in CIC-IDS-2017, 13484708 (83.06%) in CSE- CIC-IDS2018 and 3450658 (97.46%) in UNSW-NB15 are labeled as ‘Benign’ indicating huge class imbalance between the benign and attack classes.

4.1.1 CIC-IDS-2017

The dataset consists of network traffics collected from Monday to Friday at different time intervals originally extracted by the Canadian Institute for Cybersecurity for the study of network intrusion detection. CIC-IDS-2017 datasets were generated utilizing CICFlowMeter to create network flows and labelled them according to timestamps, source and target IP address, source and target ports, communication protocol and attack types. To prepare a realistic background traffic, Canadian Institute for Cybersecurity (CIC) applied the suggested B-Profile system [?] to model the abstract patterns of human interactions and generate the realistic benign background traffic from 25 real users. The 8 files combined has 79 features and 2830743 rows.

4.1.2 CSE-CIC-IDS2018

Among all the datasets used for our research, this one is the biggest in size (10 files of 6.89 GB) with more than 16 million rows and 84 feature downloaded through Kaggle. CSE- CIC-IDS2018 was a collaborative initiative between CIC and Communications Security Establishment (CSE) with the objective of creating standard benchmark datasets for intrusion detection systems to properly detect network anomalies. The dataset was generated based on different attack types such as Botnets, Brute force, DoS and Heartbleed by accessing different user profiles. The dataset is comprised of 80 features obtained through incoming network traffic and system logs with CICFlowMeter-V3 tool with proper utilization of B-profile and M-profile to identify

the statistical user behavior through the reproduction of network traffics and attack scenarios. The initiative used Slowloris and LOIC to inject HTTP DoS attacks, HOIC for DDoS, and bombarded web applications with Damn Vulnerability Web App (DVWA) to test the vulnerabilities.

4.1.3 UNSW-NB15

CIC-UNSW-NB15 or simply UNSW-NB15 augmented dataset was created with the help of IXIA PerfectStorm with up-to-date normal and abnormal network traffic. The dataset contains 9 types of attacks which the CIC-IDS-2017 and CSE- CIC-IDS2018 datasets lack. CICFlowmeter extracted new features from the network flow. The attacks in UNSW-NB15 are categorized by matching the flow records: source/destination IP, port, and protocols with the records from ground truth file. UNSW-NB15 prioritized matching Timestamp in-case of multiple records, removed flows that could not be categorized and categorized the remaining traffic as benign after malicious flows were processed.

4.2 Data Pre-Processing

The first step in data-processing is standardizing all the labels across the datasets. We mapped the labels of the datasets taking reference to CSE-CIC-IDS2018. We handled the infinite values in these datasets by replacing them with NaN values. Removing duplicates earlier would significantly reduce the size of the datasets, seeing the skewness of the datasets, we replaced the NaN values with the median values from the respective labels. And, only after replacing the NaN values with median values, we removed the duplicated values from the datasets. Finally, for the next step we took all the common labels from the datasets. We also checked and deleted the bad rows from the datasets.

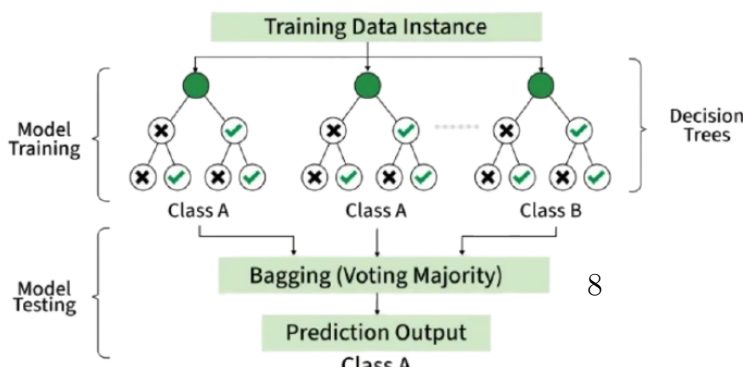
4.3 Feature Engineering

In order to maximize the intrusion detection, we mapped the attacks of similar nature. Since, the tree based models are robust in handling the class-imbalance, we have simply implemented class-weighting and avoided using SMOTE for Random Forest and XGBoost. Meanwhile, for our DNN architecture, we used RobustScaler to fit-transform the train datasets only.

4.4 Model Training

4.4.1 Random Forest

Random Forest is a robust ensemble learning model that creates many decision trees and carries out classification by taking the mode of the predictions of individual trees. During the training phase, only a random subset of features is considered at each node split. The trees in a random forest use a bootstrap-sampled version of the training data to introduce randomness in both the sampling process and feature selection, thereby minimizing correlation among the trees and reducing the variance of the ensemble without increasing bias.



The significance of tree-based models lies in their ability to classify the feature space using threshold splits without being affected by monotonic transformations of the input features. In this study, we trained

Random Forest on the full CIC-IDS-2017 dataset without scaling, despite the substantial class imbalance. As Random Forest is resistant to overfitting on high-dimensional flow-related data and shows robust-

ness to noisy features, the workflow avoids Synthetic Minority Over-Sampling Technique (SMOTE) and instead uses class weight = "balanced" to handle class imbalance.

Table 3: Hyperparameters for Random Forest

Hyperparameter	Value	Purpose
n_estimators	200	Number of trees in the forest
max_depth	10	Maximum depth of each tree
min_samples_split	2	Minimum samples to split an internal node
min_samples_leaf	2	Minimum samples for each leaf node
max_features	sqrt	Number of features at each split
bootstrap	True	Bootstrap sampling to build trees
criterion	gini	Function to measure split quality
class_weight	balanced	To automatically address the class imbalance
random_state	42	Seed for reproducibility
n_jobs	-1	Parallel Processing

4.4.2 XGBoost

eXtreme Gradient Boosting generally termed as XGBoost is a scalable gradient boosting decision tree which boost weak learners. XGBoost utilizes the technique where each tree is trained to rectify the mistakes of previous trees. It does not build trees based on parallel bagging as Random Forest does. Rather XGBoost allows trees to grow in additive way to optimize the differential loss function using gradient descent. It employs L1 and L2 regularization to control overfitting and complexity. XGBoost inhibits computational effectiveness and robustness in solving structured data and is capable to construct parallel tree to deal with the missing data.

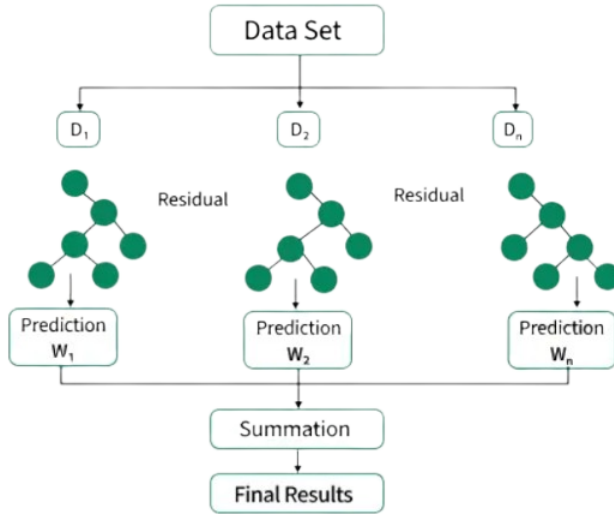


Figure 3: XGBoost Architecture

So, with its exceptional ensemble technique to compare methods of ensemble trees that differ in aggregation approach (sequential vs parallel), we have selected XGBoost along with Random Forest as our benchmark supervised learning models. Like random forest, we have implemented class-weighting to mitigate the class imbalance between the benign and malicious cases in the training set, i.e. CIC-IDS-2017.

Table 4: Hyperparameters for XGBoost

Hyperparameter	Value	Purpose
n_estimators	200	Number of boosting trees in the forest
max_depth	6	Maximum depth of each tree
learning_rate	0.1	Contribution of each tree to the final model
sub_sample	0.8	Training sub-samples to boost iteration
col_sample_bytree	0.8	Fraction of features for each tree
objective	binary:logistic	Specifies binary classification
eval_metric	logloss	Metric to calculate classification error
tree_method	hist	Histogram based training
scale_pos_weight	neg/pos	Handles class imbalance
random_state	42	Seed for reproducibility
n_jobs	-1	Parallel Processing

4.4.3 Deep Neural Network (DNN)

DNN learn feature interactions without manual feature extraction giving it an extra edge over tree-based ensemble techniques. DNN is dependent on the size/amount of data, thus shows better performance as the degree of changes in inputs rises. DNN uses hidden layers for classification. Each hidden layer utilizes ReLU activation to bring in non-linearity after each hidden layer to prevent the vanishing gradient problem of deeper models.

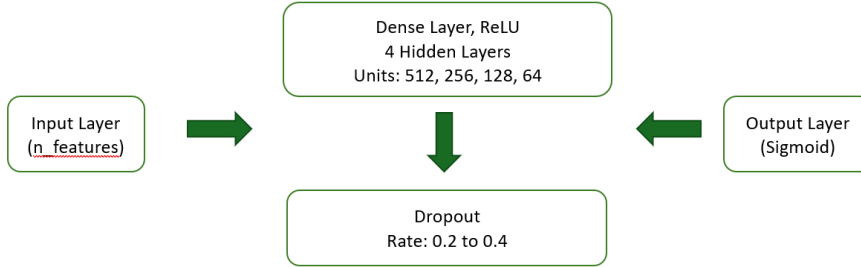


Figure 4: DNN Architecture for this study

To avoid memory crash given the size of the dataset, we tested on 20% of the CIC-IDS-2017 with tuning size of 300000. Furthermore, we have configured Optuna study which optimizes the hyperparameters of the model [?] to get maximum F1 score for the "Attack" class with n_trials=30 (30 only con-

sidering the computational cost and to prevent memory crash). We have implemented 4 hidden layers with 512, 256, 128 and 64 neurons for binary classification "benign" and "attack" in this study. We have implemented RobustScaler for scaling the datasets to mitigate the effect of strong outliers (flow duration, packet inter-arrival times). Similar to Random Forest and XGBoost which utilizes class-weight, weighted loss in DNN handles class imbalance. In consideration to the huge test dataset size, prediction and evaluation of the model has been performed in batches of max 4096. The best_dnn_params identifies the best configuration of the model specifically the number of layers, units per layer, dropout rate and the learning rate to construct the best DNN architecture. Thus, the evaluation using DNN architecture works in two phases, first finding the best configuration by training and testing the model on CIC-IDS-2017 and secondly using the best hyperparameters to evaluate the performance of the model on the test datasets CSE-CIC-IDS2018 and UNSW-NB15.

Table 5: Hyperparameters for DNN

Hyperparameter	Search Space	Purpose
n_layers	2, 4	Controls network depth, learns complex features and balances overfitting
units per layer	[512, 256, 128, 64]	Controls width of hidden layer and checks the representational capacity at each layer
learning_rate	[1e-4, 1e-2]	Controls the step size during gradient descent
dropout_rate	0.1, 0.5	Deactivate the neurons during training to prevent overfitting and overall regularizes the network
batch_size	[512, 1024, 2048, 4096]	Prevents memory crashes, training speed and gradient noise by controlling the number of samples processed

Chapter 5

Evaluation and Results

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Chapter 6

Conclusion

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References

- [Akiba et al., 2019] Akiba, T., Sano, S., Yanase, T., Ohta, T., and Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 2623–2631.
- [Cantone et al., 2024] Cantone, M., Marrocco, C., and Bria, A. (2024). Machine learning in network intrusion detection: A cross-dataset generalization study. *IEEE Access*, 12:144489–144508.
- [Chandra and Anandappa, 2025] Chandra, R. and Anandappa (2025). Real-time network intrusion detection system using deep learning: Identifying cyber attacks through anomaly pattern recognition. *Journal of Scientific Research and Technology*, pages 106–117.
- [de Carvalho Bertoli et al., 2023] de Carvalho Bertoli, G., Junior, L. A. P., Saotome, O., and dos Santos, A. L. (2023). Generalizing intrusion detection for heterogeneous networks: A stacked-unsupervised federated learning approach. *Computers & Security*, 127:103106.
- [Elangovan et al., 2026] Elangovan, R., Parthasarathy, D. D., Jawahar, M., Kaliyaperumal, P., Balusamy, B., Yogarayan, S., and Venkatesan, V. (2026). Cross-dataset temporal and semantic generalization of intrusion detection models for the future internet. *Future Internet 2026, Vol. 18, Page 194*, 18:194.
- [Hamidou and Mehdi, 2025] Hamidou, S. T. and Mehdi, A. (2025). Enhancing ids performance through a comparative analysis of random forest, xgboost, and deep neural networks. *Machine Learning with Applications*, 22:100738.
- [Kottilingal, 2024] Kottilingal, S. (2024). Deep learning based network intrusion detection system: A deep abstract networks (danets) model approach. *International Research Journal of Computer Science*, 11:539–544.
- [Miani et al., 2025] Miani, R. S., Bernardo, G. D. G., Cassales, G. W., Senger, H., and de Faria, E. R. (2025). A survey of data stream-based intrusion detection systems. *IEEE Access*, 13:72953–72983.
- [Moustafa and Slay, 2015] Moustafa, N. and Slay, J. (2015). Unsw-nb15: A comprehensive data set for network intrusion detection systems (unsw-nb15 network data set). *2015 Military Communications and Information Systems Conference, MilCIS 2015 - Proceedings*.
- [Sharafaldin et al., 2018] Sharafaldin, I., Lashkari, A. H., and Ghorbani, A. A. (2018). Toward generating a new intrusion detection dataset and intrusion traffic characterization. *International Conference on Information Systems Security and Privacy*, 2018-January:108–116.
- [ul Haq Qazi et al., 2022] ul Haq Qazi, E., Imran, M., Haider, N., Shoaib, M., and Razzak, I. (2022). An intelligent and efficient network intrusion detection system using deep learning. *Computers and Electrical Engineering*, 99.
- [Wicaksana and Umam, 2026] Wicaksana, F. A. and Umam, C. (2026). Comparative analysis of random forest and xgboost performance for network flow-based malware classification. *INOVTEK Polbeng - Seri Informatika*, 11:108–115.